



PDA Preview
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- Introducing the PDA
 - Description and applications
- Technical description
 - CPU, memory, other features
- Software development
 - Hardware and software tools
- Miscellaneous details
 - Support, licensing and launch

What is the PDA?

- A new platform, and a new peripheral
 - Software supplied as part of a PlayStation title
- Same size as a Memory card
 - Backwards compatible with Memory cards
- Additional features
 - Programmable CPU
 - LCD display
 - Infra-red communications
 - Real-time clock



Possible PDA applications

- Memory card
- Virtual pet
 - Pet characters from games
- Pocket watch / calendar / phone book
 - “Free” with games
- Multi-purpose remote control unit
- Edit game saves and exchange data
 - Portable garage
 - Player management



Hardware features (1)

- LCD display
 - 32x32 pixels, monochrome
 - Programmable refresh rate - 16/32/64Hz
- Buzzer
 - 10-bit DAC
- 5 buttons
 - 4 directional buttons + 1 “fire” button
 - All buttons can interrupt the CPU

Hardware features (2)

- Integral battery
 - Disposable button cell
 - Power saving must be done in software - be sensible
- Real-time clock
 - Can interrupt CPU once per second
 - 3 16-bit countdown timers can also interrupt

Hardware features (3)

- Infra-red communications
 - Transceiver built in to every PDA
 - IrDA capable
 - Will possibly also produce a carrier-based signal (TV remote etc)
 - Can be used when the PDA is a standalone device or when attached to a PlayStation

- ARM7TDMI-based
 - 32-bit RISC CPU
 - Clock speed programmable from 32KHz to 4MHz
 - 2 instruction sets, ARM and Thumb
 - 128K Flash RAM, 2K SRAM, 16K ROM
 - Interruptible by button pushes or real-time clock
 - Memory Management Unit

ARM instruction set



- Full 32-bit mode
- 13 general purpose registers plus PC, LR and SP, second bank of 15 for privileged mode
- Conditional execution on all instructions
 - Optionally update condition flags
- Rich instruction set
 - Multiplier with optional accumulate
 - Shifts free with many operations (MOV/ADD etc)
 - Multiple register load/store instructions

Thumb instruction set

- 16-bit instruction word, 32-bit registers
- Code typically 35% smaller than ARM code
- 8 general-purpose registers + PC, LR and SP
 - Map to ARM registers 0 to 7
 - 8 “High” registers for MOV, ADD and CMP
- Same basic instructions as ARM, but
 - Separate shift instructions
 - Conditional execution only on branches
 - Condition flags always updated

PDA memory

- 16K ROM
 - Actually Flash, but not for general use
 - Contains kernel, boot code & OSD
 - Kernel controls memory card functionality
- 128K Flash RAM
 - Contains game saves and PDA executables
 - 8K blocks mapped to address space by MMU
- 2K SRAM
 - Usable by executing programs

Developing PDA applications

- Development hardware supplied by SCEE
- Tool chains already available
 - ARM Software Development Toolkit
 - GNU tools - to be supported by SN Systems
- Additional tools provided by SCEE
 - Specifications of low-level Memory card format
 - Specifications of PDA hardware, but no libraries
 - New Memory card library for PlayStation

PDA Hardware tools

- DEX-4000
 - Standalone development box
 - Contains all PDA hardware
- Multi-ICE
 - Plugs in to PC and DEX-4000
 - Offers in-circuit debugging facilities



SDevTC development environment

- Pro-DG IDE
- 32-bit GNU tools
 - Separate compilers for ARM and Thumb
- SDevTC Windows debugger
 - To offer full PDA support
- Real-time PDA emulation

ARM development environment

- IDE
 - Project manager
 - Editor
- C compiler and assembler
 - Can produce both ARM and Thumb code
- Debugger
 - Powerful source and object-level debugger
 - ARMulator

Getting support

- Supported the same way as PlayStation
 - Developer Support hotline and e-mail
 - Developers' web site
- PDA newsgroups
 - Recommended point of contact for developers
 - Monitored by support engineers, other developers and SN Systems

Software licensing

- All licensed PlayStation developers can develop PDA applications
 - SDevTC and ARM tools supplied by SCEE
 - License fee for DEX-4000 and Multi-ICE to be confirmed
- No additional software royalties
 - Software will be seen as enhancements for PlayStation titles rather than new products

PDA launch details

- Retail price under £30
- Launch in Japan for Christmas 1998
- SCEE/SCEA territories' dates to be confirmed
 - Possible SCEE territories launch Q2 1999
 - First hardware tools available Q3 (or Q4) 1998
- Specification identical worldwide

PDA Questions?

- `scee.dev.programming.pda`
- `Dev_Support@Playstation.sony.com`
- Or ask me now